

Eenheid 5.5

Note Title

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Onbepaalde Integrals

Stewart
p402

Definitie: $\int f(x) dx = F(x) + C$

betekent $F'(x) = f(x)$ en

$\int f(x) dx$ noemt ons in onbepaalde
integraal.

Eienskappen: ① $\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$

• Tabel: Stewart
p403

• studie gids
• clickup

② $\int [f(x) - g(x)] dx = \int f(x) dx - \int g(x) dx$

③ $\int c f(x) dx = c \int f(x) dx$

3 Verskillende tipes wae:

① Die opp. tussen $f(x) = \sin x$, die x-as
en $x = -\frac{\pi}{2}$, $x = \frac{\pi}{2}$

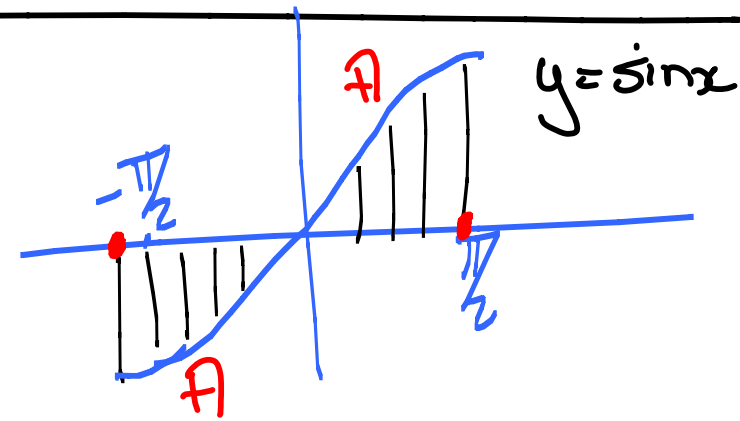
$$\int_{-\pi/2}^{\pi/2} \sin x \, dx$$

$$= - \int_{-\pi/2}^0 \sin x \, dx + \int_0^{\pi/2} \sin x \, dx$$

$$= - \left(-\cos x \right) \Big|_{-\pi/2}^0 + \left(-\cos x \right) \Big|_0^{\pi/2}$$

$$= \cos x \Big|_{-\pi/2}^0 - \cos x \Big|_0^{\pi/2}$$

$$= [\cos 0 - \cos(-\frac{\pi}{2})] - [\cos \frac{\pi}{2} - \cos 0] = 1 + 1 = 2$$



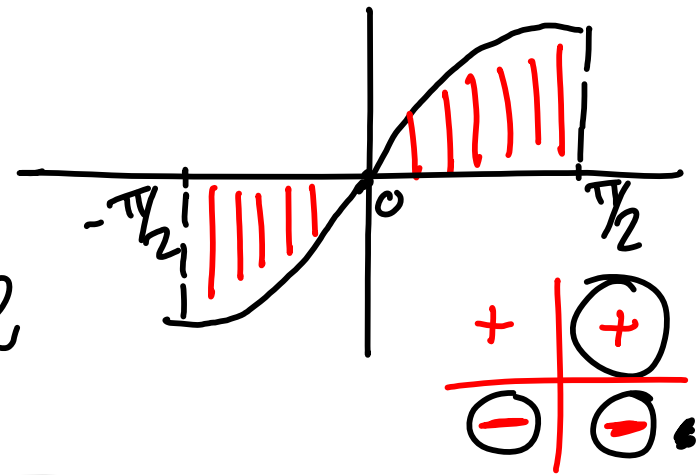
$$\int_{-\pi/2}^{\pi/2} \sin x \, dx = 2A$$

$$O_{pp} = \int_{-\pi/2}^{\pi/2} |\sin x| dx$$

$$|\sin x| = \begin{cases} \sin x & \text{as } \sin x \geq 0 \\ -\sin x & \text{as } \sin x < 0 \end{cases}$$

$$= \int_{-\pi/2}^0 (-\sin x) dx + \int_0^{\pi/2} (\sin x) dx$$

$$= -(-\cos x) \Big|_{-\pi/2}^0 + (-\cos x) \Big|_0^{\pi/2} = 2$$



② Evalueer / vind / bepaal

(Bepaalde integraal)

$$\int_{-\pi/2}^{\pi/2} \sin x dx = -\cos x \Big|_{-\pi/2}^{\pi/2}$$

$$= -\left[\cos \frac{\pi}{2} - \cos\left(-\frac{\pi}{2}\right)\right]$$

$$= -[0 - 0] = 0$$

③ Evalueer / vind / bepaal

(onbepaalde integraal)

$$\int \sin x dx = -\cos x + C$$

VOORBEELDE: Groep 1 (Mag)

$$\textcircled{1} \int x^{\textcircled{5}} dx = \frac{x^6}{6} + C$$

$$\textcircled{2} \int (1-2x)^{\textcircled{5}} dx = \left(\frac{1}{6}\right)\left(-\frac{1}{2}\right) (1-2x)^6 + C$$

Zoai
en
Toets

$$\frac{d}{dx} (1-2x)^6 = 6(1-2x)^5 \cdot (-2)$$

↓

$$\textcircled{3} * \int \sin^{\textcircled{5}} 2x \cos 2x dx$$
$$\int (f(x))^5 \times f'(x) dx$$

"Magic"

$$= \left(\frac{1}{2}\right)\left(\frac{1}{6}\right) \sin^6 2x + C$$

Toets: $\frac{d}{dx} (\sin^6 2x) = \textcircled{6} \sin^5 2x \times (\cos 2x) \times \textcircled{2}$

Groep 2

(e^x - funktie)

$$\textcircled{1} \int e^{3t} dt = \frac{1}{3} e^{3t} + C$$

$$\textcircled{2} \int \underline{x^3} e^{x^4} dx = \frac{1}{4} e^{x^4} + C$$

$\frac{d}{dx} (e^{x^4}) = e^{x^4} \times 4x^3$

$$\textcircled{3} \int e^{\tan 3x} (\sec^2 3x) dx = \frac{1}{3} e^{\tan 3x} + C$$

Watter funktie moet ik differentiëren??
 $\frac{d}{dx} (e^{\tan 3x}) = e^{\tan 3x} \times \sec^2 3x \times 3$

$$\textcircled{4} \int \underline{e^{3x} \sec^2(e^{3x})} dx = \frac{1}{3} \tan(e^{3x}) + C$$

Diff ?? $\frac{d}{dx} (\tan(e^{3x})) = \sec^2(e^{3x}) \times e^{3x} \times 3$

Groep 3

(ln x / bgtan x)

$$* \textcircled{1} \int \frac{1}{1-2x} dx = -\frac{1}{2} \ln |1-2x| + C$$

$$\frac{d}{dx} (\ln |1-2x|) = \frac{-2}{1-2x} //$$

$$\textcircled{2} \int \frac{\sin 3x}{1-\cos 3x} dx = \frac{1}{3} \ln |1-\cos 3x| + C$$

$$* \textcircled{3} \int \frac{x}{1+x^2} dx = \frac{1}{2} \ln |1+x^2| + C$$

$$\frac{d}{dx} (\ln |1+x^2|) = \frac{2x}{1+x^2}$$

$$\textcircled{4} \int \frac{3}{1+x^2} dx = 3 \operatorname{tg} \tan x + C$$

$$\frac{d}{dx} (\operatorname{tg} \tan x) = \frac{1}{1+x^2}$$

$$\frac{N3}{3} \textcircled{5} \int \frac{x}{9+x^4} dx = \int \frac{x}{9(1+\frac{x^4}{9})} dx$$

$$= \frac{1}{9} \int \frac{x \checkmark}{\left(1 + \left(\frac{x^2}{3}\right)^2\right)} = \left(\frac{3}{2}\right) \left(\frac{1}{9}\right) \text{bg tan} \left(\frac{x^2}{3}\right) + C$$

$$\frac{d}{dx} \left(\text{bg tan} \left(\frac{x^2}{3} \right) \right) = \frac{1}{1 + \left(\frac{x^2}{3}\right)^2} \times \frac{(2x)}{3}$$

Groep 4:

($\sqrt{\quad}$ / $\text{bg sin } x$)

$$\textcircled{1} \int \frac{1}{\sqrt{1-7x}} dx = -(2) \left(\frac{1}{7}\right) (1-7x)^{1/2} + C$$

$$\frac{d}{dx} (1-7x)^{1/2} = \frac{1}{2} (1-7x)^{-1/2} \times -7$$

$$\textcircled{2} \int \frac{x}{\sqrt{1-x^2}} dx = -\sqrt{1-x^2} + C$$

$$\text{Toets: } \frac{1}{2} (1-x^2)^{-1/2} \times -2x$$

$$\textcircled{3} \int \frac{1}{\sqrt{1-4x^2}} dx = \frac{1}{2} \log \sin(2x) + C$$

$$\frac{d}{dx}(\log \sin 2x) = \frac{1}{\sqrt{1-(2x)^2}} \times (2)$$

$$\textcircled{4} \int \frac{3^x}{\sqrt{1-3^x}} dx = -\frac{1}{2} \log \sqrt{1-3^x} + C$$

$$\frac{d}{dx}(\sqrt{1-3^x}) = \left(\frac{1}{2}\right)(1-3^x)^{-1/2} \times -3^x \ln 3$$

$$\textcircled{5} \int \frac{e^{3x}}{\sqrt{1-e^{6x}}} dx = \frac{1}{3} \log \sin(e^{3x}) + C$$

$$\frac{d}{dx}(\log \sin(e^{3x})) = \frac{1}{\sqrt{1-(e^{3x})^2}} \times e^{3x} \times 3$$

Gemengde Oefening Eenheid 5.5

① $\int \frac{x^4}{1-x^5} dx$
ln? / bg tan

$$= -\frac{1}{5} \ln |1-x^5| + C$$

Toets: $\frac{5x^4}{1-x^5}$

② $\int \frac{x^4}{1+x^{10}} dx$
ln / bg tan

$$= \int \frac{x^4}{1+(x^5)^2} dx = \frac{1}{5} \text{bg tan}(x^5) + k$$

Toets?

$$\textcircled{3} \int \frac{1+x^5}{x^4} dx = \int \left(\frac{1}{x^4} + \frac{x^5}{x^4} \right) dx$$

(Eenheid 5.5)

$$= \int \frac{1}{x^4} dx + \int x dx = \left(\frac{1}{-3} \right) x^{-3} + \frac{x^2}{2} + C$$

$\neq x^{-4}$

$$\textcircled{4} \int \left[\frac{\sec^2 x}{\sqrt{4+\tan x}} \right] dx = 2(4+\tan x)^{\frac{1}{2}} + C$$

Toets!!

$$\textcircled{5} \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx = 2e^{\sqrt{x}} + k$$

Toets: $\frac{d}{dx}(e^{\sqrt{x}}) = e^{\sqrt{x}} \times \frac{1}{2\sqrt{x}}$

$$\textcircled{6} * \int \frac{x^5}{\sqrt{16-x^6}} dx = -(2)\left(\frac{1}{6}\right)(16-x^6)^{\frac{1}{2}} + k$$

Toets: $\frac{d}{dx}(16-x^6)^{\frac{1}{2}} = \frac{1}{2\sqrt{16-x^6}} \times -6x^5$

⑦ $\int \frac{x^2}{\sqrt{16-x^6}} dx$

Is it $\sqrt{\text{something}}$ of trigonometry?

$$= \int \frac{x^2}{\sqrt{16 - (x^3)^2}} dx = \int \frac{x^2}{\sqrt{16} \sqrt{1 - \left(\frac{x^3}{4}\right)^2}} dx$$

$$= \frac{1}{\sqrt{16}} \int \frac{x^2}{\sqrt{1 - \left(\frac{x^3}{4}\right)^2}} dx = \frac{1}{\sqrt{16}} \cdot \frac{4}{3} \cdot \frac{1}{3} \log \sin\left(\frac{x^3}{4}\right) + C$$

$$= \frac{1}{3} \log \sin\left(\frac{x^3}{4}\right) + C$$

Tools: $\frac{d}{dx} \left(\log \sin\left(\frac{x^3}{4}\right) \right) = \frac{1}{\sqrt{1 - \left(\frac{x^3}{4}\right)^2}} \times \frac{1}{4} \times 3x^2$

$$\textcircled{8} \int \frac{\sin bx}{(1 + \cos bx)^3} dx = \int \cancel{4} \sin bx \times (1 + \cos bx)^{\cancel{-3}} dx$$

$$= \left(\frac{1}{2}\right) \left(\frac{1}{6}\right) (1 + \cos bx)^{-2} + C$$

Tests: $\cancel{-2} (1 + \cos bx)^{-3} \times \cancel{-} \sin(bx) \times \cancel{6}$

$$\textcircled{9} \int \boxed{8 \operatorname{cosec} 5x} \operatorname{cosec} 5x \cot 5x dx$$

$$= -\left(\frac{1}{5}\right) \left(\frac{1}{\ln 8}\right) 8 \operatorname{cosec} 5x + k$$

Tests: $\frac{d}{dx} (8 \operatorname{cosec} 5x) = 8 \operatorname{cosec} 5x \ln 8 \times -\operatorname{cosec} 5x \cot 5x \times 5$

*10) $\int \tan 2x \, dx = \int \frac{\sin 2x}{\cos 2x} \, dx$
 $= -\frac{1}{2} \ln |\cos 2x| + C$

*11) $\int \frac{\sinh(\frac{1}{x})}{x^2} \, dx = -\cosh(\frac{1}{x}) + C$
 Toets: $\frac{d}{dx} \left(\cosh(\frac{1}{x}) \right) = \sinh(\frac{1}{x}) \cdot \frac{-1}{x^2}$

12) $\int x \cosh(x^2) \cdot 3^{\sinh(x^2)} \, dx$
 $= \left(\frac{1}{2}\right) \left(\frac{1}{\ln 3}\right) 3^{\sinh(x^2)} + k$

$$13). \int \frac{\operatorname{cosec}^2(\ln x)}{x} dx$$

$$= -\cot |\ln x| + C$$

$$14) \int x e^{x^2} \sec(e^{x^2}) \tan(e^{x^2}) dx$$

$$= \frac{1}{2} \sec(e^{x^2}) + k$$

$$\begin{aligned} 15. \quad \int (1-x^3)^2 dx &= \int (1-x^3)(1-x^3) dx \\ &= \int (1 - 2x^3 + x^6) dx \\ &= x - \frac{2x^4}{4} + \frac{1}{7}x^7 + C \end{aligned}$$

$$\begin{aligned} 16. \quad \int \frac{2+x^4}{x^2} dx &= \int \frac{2}{x^2} dx + \int x^2 dx \\ &= -2x^{-1} + \frac{x^3}{3} + C \end{aligned}$$

Extra Voorbeelde :

$$\begin{aligned}\textcircled{1} \int x^3 \sqrt{x^4+5} \, dx &= \int x^3 (x^4+5)^{1/2} \, dx \\ &= \left(\frac{1}{4}\right) \left(\frac{2}{3}\right) (x^4+5)^{3/2} + C\end{aligned}$$

$$\begin{aligned}\textcircled{2} \int \frac{2+x}{\sqrt{1-x^2}} \, dx &= \int \frac{2}{\sqrt{1-x^2}} \, dx + \int \frac{x}{\sqrt{1-x^2}} \, dx \\ &= 2 \log \sin x - \left(\frac{1}{2}\right)(2)(1-x^2)^{1/2} + C\end{aligned}$$

$$\textcircled{3} \int \frac{t^9 \ln(1+t^{10})}{1+t^{10}} \, dt = \left(\frac{1}{2}\right) \left(\frac{1}{10}\right) [\ln(1+t^{10})]^2 + C$$

$$\begin{aligned} \textcircled{4} \quad \int \frac{2x+1}{1+4x^2} dx &= \int \frac{2x}{1+4x^2} dx + \int \frac{1}{1+4x^2} dx \\ &= (2)\left(\frac{1}{8}\right) \ln |1+4x^2| + \frac{1}{2} \operatorname{tg} \tan^{-1}(2x) + C \end{aligned}$$